FarmConnect AI Implementation Speech

Generated 2026-05-23

Good [morning/afternoon], everyone. Today I will explain how AI is implemented in the FarmConnect project and how it delivers practical value to producers and buyers.

At a high level, FarmConnect uses a dedicated AI microservice. The frontend does not talk to AI directly. Instead, the web app calls the main API, and the API calls the AI service. This gives us a secure gateway for validation, logging, and fallbacks.

The AI service is a Python FastAPI application. It exposes three core endpoints: price recommendation, demand forecasting, and a chatbot assistant. Each endpoint returns not just an answer, but also a confidence score and explainable factors, so the output is transparent and easy to trust.

Let us start with price recommendation. The web app requests a recommendation through the API. The API enriches the request with real product data such as current price, stock level, and recent orders. The AI service then computes a recommended price. The response includes the recommended value, confidence, an explanation, and a list of factors that influenced the result. If the AI service is unavailable, the API uses a conservative fallback, typically the current price or a cost-plus margin value.

Next is demand forecasting. The flow is similar. The API adds context like category, active listings, and stock when available. The AI service forecasts demand for a given horizon in days and returns confidence plus explanation. If AI cannot respond, the API produces a basic forecast from historical averages, ensuring the user still receives a useful estimate.

The third capability is the chatbot assistant. The web app sends the user message to the API, which relays it to the AI service. The response includes an answer, an intent label, confidence, and follow-up suggestions. If the AI service is down, a safe fallback response is returned so the UI never blocks.

Explainability is built into every AI response. Each recommendation and forecast includes factors with impact and weight, plus a human-readable explanation. The UI clearly shows the confidence score and whether a fallback was used. This prevents blind trust and supports better decision making.

Reliability is handled through timeouts and fallbacks. AI calls are time-limited to keep the system responsive. When timeouts occur, the API logs the event and returns a fallback response. This design ensures AI adds value without becoming a single point of failure.

Finally, it is important to note that the current AI logic is heuristic-based. It is designed to provide practical guidance today while leaving room for growth. The roadmap aims to evolve these features into data-driven, measurable ML models with stronger pipelines.

In summary, FarmConnect uses AI as a trusted assistant: transparent, explainable, and resilient. It enhances pricing, forecasting, and support without compromising stability. Thank you, and I am happy to take questions.